

Competition: IoTrix 2.0

(Track A – IoT Product & System Architecture)

Elder Guard AI

**“An AI-Powered IoT-Based Personalized Elderly Health Monitoring
Emergency Assistance, and Hospital QR Summary System”**



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4. Team Details and Proposed Module Ownership

Member	Primary Module / Component	Main Technical Responsibilities	Expected Deliverables
Kathirsan	IoT Monitoring & Wearable Communication Module	Sensor data acquisition, wearable prototype, BLE communication, movement/activity monitoring, fall-detection logic, data validation, communication between wearable and mobile application	Working wearable prototype, sensor-data collection component, BLE communication, fall-detection component, device integration, technical documentation and test evidence
Abisegan	Backend API & Risk Engine Module	FastAPI backend, MySQL database design, real-time risk assessment logic (Normal, Warning, Critical), and push alert dispatch.	FastAPI service, MySQL database schema, real-time risk scoring engine, push notification integration, and API documentation.
Pavithar	Dual-Mode Mobile Application Module	Flutter app with Elderly Mode (vitals, medication/food reminders, SOS, QR code generator) and Caregiver Mode (remote vital monitoring, push notifications).	Dual-mode mobile application, interactive dashboard UI, medication/food reminder engine, patient QR generator, and caregiver alert receiver.

Saruhasan	Hospital QR Web Dashboard & Print System	Web dashboard for clinic staff, tokenized QR scanning verification, and print-optimized CSS view for 1-click Ctrl+P medical summary printing.	Zero-login hospital web application, QR scanning decoder, 1-page print stylesheet, and dashboard integration documentation.
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5. Introduction and Problem Definition

5.1 Background and Context

The increasing number of elderly people living independently creates significant challenges in ensuring their safety, health monitoring, medication adherence, and timely access to assistance during emergencies. Elderly individuals who live alone may experience sudden health abnormalities, falls, medication-related issues, reduced mobility, or other unexpected incidents without having a caregiver continuously available to observe their condition. In such situations, delays in identifying an abnormal condition and notifying an appropriate person can increase the potential consequences of the incident.

Traditional elderly-care practices often depend on periodic visits by family members, caregivers, or healthcare professionals. Although these approaches provide important human support, continuous observation of an elderly person's health and daily activities is generally difficult to achieve. Wearable and mobile technologies provide an opportunity to collect health and activity information continuously and communicate this information to software systems for further analysis.

However, simply collecting health measurements is not sufficient. Health measurements can vary between individuals, and an abnormal value for one person may not necessarily represent the same level of risk for another person. Therefore, an effective elderly-care system should consider individual health characteristics, historical measurements, activity patterns, medication information, and previously observed incidents when assessing potential risks.

ElderGuard AI is proposed for the IoTrix 2.0 competition under Track C (IoT Product & System Architecture). It is an AI-powered IoT-based elderly health monitoring, emergency assistance, and hospital integration system. The solution pairs an ESP32 wearable device with a dual-mode Flutter mobile application (Elderly Mode and Caregiver Mode) communicating over BLE. For clinical encounters, the elderly phone displays a unique tokenized QR code that hospital staff can scan without logging in, opening a zero-login web dashboard designed for 1-click Ctrl+P printing of a concise 1-page health summary for consulting doctors.

5.2 Target Users and Stakeholders

The primary target users and read-only actors of the system are:

1. Elderly Person (Authenticated User):

Elderly individuals who wear the ESP32 device and interact with the Elderly Mode of the Flutter application. They can view simplified vital metrics, receive medication and food reminders, trigger manual SOS requests, and display their patient QR code when visiting clinics.

2. Family Caregiver (Authenticated User):

Family members or caregivers who use the Caregiver Mode of the Flutter application to remotely monitor real-time health data, track risk status, and receive instant push notifications during warning or critical health events.

3. Hospital Clinic Staff (Read-Only Unauthenticated Actor):

Hospital receptionists or clinic staff who scan the elderly user's phone QR code. They gain zero-login access to a web dashboard that renders a print-optimized 1-page health summary (Ctrl+P) for the consulting doctor.

5.3 Current Situation

In many elderly-care situations, monitoring is primarily performed through direct human interaction, scheduled visits, telephone communication, or individual health-monitoring devices. These approaches can provide valuable information but may not provide continuous monitoring or automated analysis of multiple health and activity factors.

Wearable devices can collect measurements such as heart rate, blood oxygen saturation, body temperature, and movement information. However, raw measurements alone do not necessarily provide sufficient context to determine whether an elderly person is experiencing an unusual or potentially dangerous situation.

Similarly, medication reminders can inform a person that medication should be taken, but a reminder alone does not necessarily provide a mechanism for monitoring whether the medication was taken, associating medication adherence information with subsequent health patterns, or notifying an authorized caregiver when important events occur.

Consequently, there is a need for a more integrated approach that connects wearable monitoring, mobile interaction, historical health information, medication management, intelligent analysis, and emergency response.

5.4 Problem Definition

The primary problem addressed by this project is the difficulty of providing continuous, personalized, and timely monitoring and emergency assistance for elderly individuals who live independently without a permanently available caregiver.

The problem is not limited to the absence of health-monitoring devices. The key challenge is the lack of an integrated mechanism capable of collecting relevant health and activity information, understanding the individual's normal patterns, identifying significant deviations, considering contextual information such as medication adherence and previous incidents, assessing the severity of a situation, and communicating appropriate alerts to authorized people.

For example, a single abnormal heart-rate measurement may not provide enough information to determine whether an emergency is occurring. However, a combination of abnormal vital signs, a detected fall, reduced movement, and failure to respond to an alert may indicate a substantially higher level of risk. The system therefore needs to combine multiple sources of information and apply a structured risk-assessment process rather than relying only on isolated threshold-based measurements.

5.5 Limitations and Difficulties in the Current Situation

Several difficulties can affect the effective monitoring of elderly individuals living independently:

- Continuous human monitoring is difficult because family members and caregivers cannot remain with the elderly person at all times.
- Periodic health checks may fail to identify unexpected incidents occurring between monitoring sessions.
- Raw wearable measurements may not provide sufficient contextual information for understanding the individual's condition.
- Generic health thresholds may not adequately represent differences between individual elderly persons.
- Medication reminders alone do not guarantee medication adherence or provide sufficient information about missed medication events.
- Falls or other emergency situations may remain unnoticed when an elderly person is alone.
- Family members or caregivers may not receive timely information about important health or emergency events.
- Historical health measurements and previous incidents may not be effectively combined for identifying longer-term trends.
- Multiple health and activity indicators may need to be considered together to distinguish normal variations from potentially significant abnormalities.

5.6 Motivation for the Proposed Solution

The motivation for developing ElderGuard AI is to provide an integrated technological approach that can support elderly individuals while allowing them to maintain a greater degree of independence.

The proposed system will combine an IoT-based wearable monitoring layer with a mobile application, backend services, personalized health profiles, AI-assisted analysis, risk assessment, medication management, and emergency notification mechanisms.

Rather than treating every elderly person using identical predefined thresholds, the system will use individual health information and doctor-defined parameters to support personalized assessment. Historical measurements and activity information can also be analysed to identify unusual patterns or gradual changes.

The system is intended to assist, rather than replace, healthcare professionals, caregivers, or emergency services. Its purpose is to provide timely information and structured alerts that can support appropriate human intervention when a potentially significant event is detected.

5.7 Significance and Practical Relevance

The proposed project has practical relevance because it addresses the challenges associated with supporting elderly individuals who live independently. Early identification of abnormal conditions and timely communication with authorized caregivers may help reduce delays in responding to potentially serious situations.

The integration of wearable sensing, mobile computing, AI-based analysis, and emergency response also provides significant technical value as a Software Engineering Mini Project. The system requires the design and integration of multiple components, including IoT communication, mobile application development, backend services, data management, personalized risk assessment, event detection, notification workflows, security mechanisms, and system-level testing.

From a user perspective, the system aims to provide elderly individuals with a simple monitoring and assistance interface while giving authorized family members, caregivers, and healthcare professionals access to relevant information according to their responsibilities. From a technical perspective, the project provides opportunities to investigate how heterogeneous health and activity data can be collected, processed, analysed, and transformed into actionable risk information within a mobile-centered software ecosystem.

Therefore, ElderGuard AI addresses a real-world elderly-care problem while providing a technically meaningful platform for applying mobile application development, IoT communication, intelligent data analysis, secure software engineering, and integrated emergency-response mechanisms.

6. Existing Solutions and Gap Analysis

6.1 Overview of Existing Solutions

Several commercial products and research systems have been developed to support elderly safety, health monitoring, fall detection, and emergency assistance. Existing solutions demonstrate the practical feasibility of wearable sensing, activity monitoring, automated fall detection, emergency communication, and health-data collection. However, these capabilities are often distributed across different products or focused on specific aspects of elderly care.

Research has also investigated the use of wearable sensors and Internet of Things (IoT) technologies to support independent living among older adults. A systematic review of wearable sensors and IoT-based monitoring applications identified their potential for supporting older adults living independently while also highlighting technical and practical challenges associated with these systems. Another systematic review of IoT applications for elderly health and safety reported the use of IoT technologies for monitoring older adults and evaluated systems using measures such as accuracy, sensitivity, and specificity.

The following existing solutions are particularly relevant to the proposed ElderGuard AI system.

6.2 Apple Watch and Apple Health

Apple Watch provides several health, safety, and emergency-related capabilities. Its Fall Detection feature uses motion sensors and algorithms to detect hard falls. When a hard fall is detected, the device can alert the user and, under configured conditions, automatically contact emergency services if the user remains immobile. Emergency contacts can also receive information including the user's location. Apple also provides Medical ID functionality for storing important medical information that can be accessed by first responders under appropriate conditions.

Strengths:

- Provides wearable-based health and safety monitoring.
- Supports automatic fall detection.
- Can assist with emergency communication.
- Provides emergency-contact and location capabilities.
- Integrates health and safety information within a mature consumer ecosystem.

Limitations in relation to the proposed project:

- The solution is primarily designed as a commercial consumer ecosystem rather than as a customizable elderly-care platform for a university-specific prototype.
- The project does not provide direct control over the underlying commercial hardware and algorithms.
- The proposed ElderGuard AI system intends to investigate its own wearable-data acquisition, personalized risk assessment, and decision-making workflow.
- ElderGuard AI focuses specifically on combining health measurements, medication adherence, historical information, and multiple risk indicators into a unified elderly-care workflow.

6.3 Samsung Galaxy Watch

Samsung Galaxy Watch devices also provide health and safety functionality, including hard-fall detection and emergency SOS capabilities. Samsung documentation indicates that supported Galaxy Watch devices can detect hard falls and send SOS information to configured emergency contacts. The system can also share location information when configured appropriately. Samsung explicitly notes that fall detection cannot detect every fall and that network connectivity and smartphone pairing may be required for certain emergency functions.

Strengths:

- Wearable-based fall detection.
- Emergency SOS functionality.
- Emergency-contact configuration.
- Location sharing.
- Integration with a smartphone.

Limitations in relation to the proposed project:

- The system is primarily a commercial wearable ecosystem with predefined functionality.
- Fall detection is focused on detecting falls rather than providing a broader personalized elderly-care risk-assessment workflow.

- The proposed ElderGuard AI system aims to combine multiple information sources, including current health data, historical measurements, medication adherence, activity information, and individual health profiles.
- ElderGuard AI will provide an integrated research prototype in which the group can control and evaluate the communication and risk-assessment components.

6.4 Philips Lifeline

Philips Lifeline provides dedicated medical-alert and fall-detection services for older adults. Its fall-detection systems use sensors such as accelerometers and barometric sensors together with algorithms to identify potential falls. When a fall is detected, the system can establish communication with a response center, allowing trained personnel to assess the situation and arrange appropriate assistance when required.

Strengths:

- Specifically designed for elderly safety and medical-alert scenarios.
- Supports automatic fall detection.
- Provides communication with trained response personnel.
- Can support emergency assistance even when the user cannot manually activate a help button.

Limitations in relation to the proposed project:

- The primary focus is emergency/medical-alert response rather than comprehensive personalized health analysis.
- The system is a commercial service involving dedicated response infrastructure.
- It does not represent a student-controlled platform in which wearable data, AI analysis, medication adherence, patient-specific profiles, and historical trends can be integrated and experimentally evaluated as a single software system.
- The proposed ElderGuard AI project aims to combine monitoring, analysis, medication management, personalized risk assessment, and caregiver/doctor communication.

6.5 Academic IoT and Wearable Monitoring Systems

Academic research has investigated IoT-based monitoring systems for elderly people living independently. A systematic review of wearable sensors and IoT-based applications identified applications for monitoring older adults and supporting independent living, while also highlighting challenges associated with wearable and IoT technologies.

Research has also examined fall detection using wearable, non-wearable, and hybrid sensor configurations. A recent systematic review evaluated fall-detection systems according to measures including accuracy, precision, sensitivity, specificity, F1-score, and computational performance. This demonstrates that fall detection is an established research area, rather than a completely new technical problem.

The existing research demonstrates the feasibility of collecting sensor information and applying machine-learning or rule-based approaches to detect abnormal events. However, the literature also indicates challenges such as limited fall-event data and the difficulty of achieving reliable detection

across different users and environments. Recent research specifically identifies limited training data and the rarity of real fall events as important challenges for AI-based fall detection.

Strengths of existing research:

- Demonstrates the feasibility of wearable and IoT-based monitoring.
- Provides established approaches for activity and fall detection.
- Demonstrates the use of machine learning and other algorithms for health-related pattern analysis.
- Provides measurable evaluation methods for monitoring and fall-detection systems.

Limitations and opportunities:

- Many studies focus on individual technical problems such as fall detection rather than a complete elderly-care workflow.
- Different systems may focus on specific sensor types or particular monitoring environments.
- Combining multiple sources of information into a personalized risk-assessment mechanism remains technically challenging.
- Real-world AI evaluation can be affected by limited datasets and the rarity of actual emergency events.

6.6 AI-Based Fall Detection Research

Machine learning and deep learning have increasingly been investigated for fall detection and fall-risk prediction. Recent reviews indicate that AI-based approaches can be used for detecting falls and investigating future fall risk, but also identify challenges related to data availability, model reliability, and generalization.

Existing research therefore provides an important technical foundation for ElderGuard AI. However, the proposed project will not attempt to claim that it has invented fall detection or AI-based health monitoring. Instead, the project will investigate how established sensing and AI techniques can be integrated into a personalized elderly-care system and evaluated within the project's available resources.

6.7 Identified Gap

Based on the reviewed solutions, the following opportunities have been identified:

1. **Integrated elderly-care workflow:** Existing products commonly specialize in particular capabilities such as wearable health monitoring or emergency/fall detection. ElderGuard AI aims to integrate monitoring, medication management, risk assessment, and emergency assistance within one system.
2. **Personalized risk assessment:** Instead of relying only on general thresholds, ElderGuard AI will incorporate doctor-defined patient information and individual baseline measurements when assessing risk.
3. **Multi-factor analysis:** The proposed system will consider combinations of current measurements, historical health information, activity patterns, medication adherence, and detected events rather than relying solely on a single measurement.

4. **Medication and health-pattern association:** Medication adherence information will be retained as contextual information that can be analysed alongside later health measurements. The system will identify patterns or associations rather than claiming that a missed medication directly causes a medical condition.
5. **Graduated emergency response:** ElderGuard AI will classify situations into different risk levels and use the classification to determine an appropriate response, ranging from continued monitoring to caregiver or doctor notification.
6. **Mobile-centered IoT integration:** The proposed system will use the mobile application as an important gateway between the wearable device, backend services, AI analysis, and authorized users. This provides a significant Mobile App Development component for the project.
7. **Student-controlled technical implementation:** Unlike relying entirely on a commercial ecosystem, the project will implement and evaluate the communication, data-processing, risk-assessment, and emergency-response components as part of an integrated prototype.

6.8 Proposed Contribution

ElderGuard AI therefore aims to provide meaningful technical value through **integration, personalization, and decision support**, rather than claiming research-level novelty.

The proposed system will combine:

IoT wearable sensing → Mobile data collection → Backend processing → Personalized AI-assisted analysis → Risk assessment → Emergency decision-making → Caregiver/doctor notification → Historical monitoring

The system will also provide measurable evaluation of the implemented components, including sensor communication, fall/event detection, risk-classification performance, notification response time, medication adherence tracking, and overall system reliability.

The project is intended as an academic prototype and decision-support system. It will not be presented as a replacement for professional medical diagnosis, clinical judgment, or emergency services.

7. Project Aim, Objectives and Scope

7.1 Aim

The aim of **ElderGuard AI** is to construct an IoT-based, dual-mode mobile health monitoring system for IoTrix 2.0 Track C. The system collects vitals via an ESP32 wearable over BLE, performs real-time risk assessment via a FastAPI backend, dispatches push alerts to caregivers, and provides a zero-login hospital web dashboard accessible via QR code for 1-click Ctrl+P medical summary printing.

7.2 Objectives

The following objectives are defined to achieve the overall aim of the project:

1. **To construct an ESP32 wearable prototype** that continuously gathers heart rate, SpO2, temperature, and motion/fall data and transmits it to the mobile application via BLE.

2. **To develop a dual-mode Flutter mobile application** supporting Elderly Mode (vitals display, medication/food reminders, manual SOS, and patient QR display) and Caregiver Mode (remote vital tracking and push notifications).
3. **To deploy a FastAPI backend and MySQL database** to process incoming sensor streams, maintain historical records, and evaluate real-time health risk levels (Normal, Warning, Critical).
4. **To implement a zero-login Hospital Web Dashboard** accessible by scanning the patient's mobile QR code, allowing clinic staff to print a 1-page health summary (Ctrl+P) for the consulting doctor.

7.3 Scope

7.3.1 In Scope

The following components and functions are included within the scope of the ElderGuard AI Mini Project.

A. IoT and Wearable Monitoring

- Development of a wearable monitoring prototype using suitable microcontroller and sensor hardware.
- Collection of selected health measurements.
- Collection of movement/activity information.
- Basic fall/event detection using suitable sensor data.
- Communication between the wearable device and the mobile application using an appropriate wireless communication mechanism, such as BLE.
- Validation and preprocessing of received sensor data.

B. Elderly Mobile Application

- Elderly user authentication and secure access.
- Simple elderly-oriented user interface.
- Health monitoring dashboard.
- Display of selected current health measurements.
- Display of current risk status.
- Wearable-device connection and data reception.
- Medication reminders.
- Medication confirmation and missed-medication recording.
- Notifications for warnings and emergency events.
- Emergency interaction and response confirmation where applicable.
- Basic local handling of selected emergency conditions when supported by the device and application environment.

C. Backend and Data Management

- User and role management.
- Elderly profile management.
- Doctor and caregiver association with authorized elderly users.
- Wearable-device registration and management.

- Storage of health measurements.
- Storage of historical health and activity information.
- Medication schedule and medication-history management.
- Storage of risk assessments and emergency events.
- RESTful or equivalent APIs for communication between the mobile application and backend.
- Secure database management.

D. Personalized Health Profile

Authorized healthcare users will be able to configure relevant information for an elderly person, including:

- Basic demographic information required by the system.
- Individual baseline health ranges or parameters.
- Doctor-defined risk-related thresholds.
- Relevant medication information.
- Relevant emergency configuration information.

These parameters will be used as contextual information during risk assessment.

E. AI-Assisted Health Analysis

The project will implement an AI-assisted or data-driven analysis component for:

- Detection of unusual patterns in selected health measurements.
- Comparison of current measurements with individual historical patterns.
- Basic activity-pattern analysis.
- Health-trend analysis.
- Identification of potentially significant combinations of events.
- Personalized risk assessment using individual profile information and available historical data.

The AI component will be evaluated using appropriate performance measures based on the available dataset and/or controlled experimental data.

F. Risk Assessment and Emergency Response

The system will use predefined risk categories:

- **Normal**
- **Warning**
- **High Risk**
- **Critical**

The risk-assessment mechanism will consider relevant combinations of health, activity, medication, historical, and event information.

For critical events, the system will generate alerts for configured and authorized recipients such as family members, caregivers, or doctors.

G. Doctor and Caregiver Functions

Authorized doctors and caregivers will be able to:

- View assigned elderly users.
- View relevant current health information.
- View historical health trends.
- Review medication adherence.
- Review abnormal events.
- Review risk assessments.
- Receive appropriate alerts.
- Manage information according to their authorized access level.

H. Security and Privacy

The system will include:

- User authentication.
- Role-based authorization.
- Secure communication between application and backend.
- Access restrictions for health-related information.
- Appropriate protection of stored user information.
- Basic auditability of important health and emergency-related events.

I. Supported Platforms and Devices

The initial prototype will target:

- **Android mobile devices** for the elderly mobile application and related mobile functionality.
- A suitable **ESP32-class microcontroller and selected sensors** for the wearable/monitoring prototype.
- A backend server environment for application services and AI processing.
- A relational database for persistent system data.

The exact sensor models and hardware configuration will be finalized according to availability, cost, reliability, and project feasibility.

7.3.2 Out of Scope

To maintain a realistic Mini Project scope, the following functions and activities are explicitly excluded:

1. **Medical diagnosis:** The system will not diagnose diseases or replace a qualified doctor's clinical judgment.
2. **Autonomous medical treatment:** The system will not prescribe, modify, or administer medication automatically.
3. **Medical-grade device certification:** The prototype will not be presented as a certified medical device and will not undergo formal clinical certification.

4. **Guaranteed emergency-service dispatch:** Direct integration with ambulance, hospital, police, or other emergency-service infrastructure will not be assumed unless an officially accessible and technically feasible service is available. The project will primarily implement configurable emergency notification and escalation mechanisms.
5. **Blood-glucose monitoring requiring specialized medical hardware:** Blood-glucose measurement will not be included unless suitable and safe hardware is available within the project resources.
6. **Clinical ECG diagnosis:** ECG-based clinical diagnosis and interpretation will not be included in the Mini Project.
7. **Full hospital information-system integration:** Integration with hospital electronic medical record or hospital management systems is outside the scope.
8. **Large-scale commercial deployment:** The project will produce an academic prototype rather than a production-scale commercial healthcare platform.
9. **24/7 professional monitoring centre:** Establishing and operating a dedicated medical emergency response centre is outside the scope.
10. **Guaranteed detection of every fall or medical emergency:** The system will be evaluated as a technical prototype, and limitations such as sensor errors, connectivity problems, false positives, and false negatives will be documented.
11. **Advanced clinical prediction of specific diseases:** The AI component will focus on anomaly detection, trend analysis, and risk assessment rather than predicting or diagnosing specific diseases.
12. **Unlimited sensor integration:** Only selected sensors required to demonstrate the project's objectives will be implemented.
13. **iOS application development:** The initial Mini Project prototype will target Android devices. iOS support is outside the primary scope unless resources and project time permit.

7.4 Scope Boundaries and Feasibility

The scope has been intentionally defined to balance technical depth with the available Mini Project development period. The project includes sufficient work across IoT communication, mobile application development, backend services, AI-assisted analysis, personalized risk assessment, medication management, emergency response, security, and testing.

At the same time, clinical diagnosis, medical treatment, certified medical-device development, direct emergency-service infrastructure integration, and large-scale commercial deployment are excluded. These boundaries allow the team to focus on implementing and evaluating the core technical contributions of ElderGuard AI within a realistic academic project environment.

8. Requirements Analysis

The requirements of ElderGuard AI are derived from the identified problem, project objectives, proposed system architecture, and needs of the elderly person, doctor, caregiver, and system administrator. The requirements are divided into functional and non-functional requirements.

8.1 Functional Requirements

FR01 – User Registration and Authentication

The mobile application shall support secure authentication for exactly two user roles: Elderly Person and Family Caregiver..

The system shall support the following primary roles:

- Elderly person
- Family member/caregiver

The system shall validate user credentials before providing access to protected functions.

FR02 – Role-Based Authorization

- The application shall restrict features according to the active mode (Elderly Mode vs. Caregiver Mode) upon successful authentication.

FR03 – Elderly Profile Management

The system shall allow authorized users to create and manage an elderly person's profile.

The profile shall contain relevant information required for personalized monitoring and risk assessment.

FR04 – Clinical QR Code Generation

The mobile application shall generate a unique, tokenized patient QR code on the elderly user's phone. This QR code encodes a secure access token linking to the patient's recent vital sign history and health summary.

FR05 – Wearable Device Registration

The system shall allow an authorized user or administrator to register and associate a supported wearable monitoring device with an elderly user's account.

The system shall maintain device identification and association information.

FR06 – Sensor Data Acquisition

The wearable prototype shall collect selected health and activity information using supported sensors.

Depending on the final hardware configuration, the system may collect:

- Heart rate
- SpO₂
- Body temperature
- Movement/acceleration
- Activity-related information

The system shall validate received measurements before transmitting or storing them.

FR07 – Wearable-to-Mobile Communication

The system shall establish communication between the supported wearable device and the elderly user's mobile application using an appropriate wireless communication mechanism such as Bluetooth Low Energy (BLE).

The mobile application shall receive sensor measurements transmitted by the wearable device.

The system shall identify connection failures or invalid data transmissions where possible.

FR08 – Real-Time Health Data Display

The elderly mobile application shall display relevant received health measurements in an understandable format.

The dashboard may include:

- Heart rate
- SpO₂
- Temperature
- Activity status
- Current risk status

The system shall indicate when recent sensor data is unavailable or the wearable connection is interrupted.

FR09 – Health Data Storage

The backend shall store validated health measurements together with relevant timestamps and user/device information.

The system shall maintain historical measurements for subsequent analysis and reporting.

FR10 – Historical Health Analysis

The system shall retrieve historical health measurements for an authorized elderly user.

The system shall support analysis of health and activity trends over appropriate periods such as daily, weekly, and monthly intervals.

FR11 – Anomaly Detection

The AI-assisted analysis component shall identify unusual patterns in selected health and activity measurements.

The analysis shall consider relevant current measurements and historical information.

The system shall record detected anomalies together with relevant timestamps and associated information.

FR12 – Personalized Risk Assessment

The system shall calculate a risk level for the elderly user using relevant information such as:

- Current health measurements
- Individual baseline parameters
- Historical measurements
- Activity information
- Medication adherence
- Previous abnormal events
- Detected falls or other significant events

The system shall classify the assessed condition into predefined categories such as:

- Normal
- Warning
- High Risk
- Critical

The system shall store the resulting risk assessment for authorized review.

FR13 – Activity and Fall/Event Detection

The system shall analyse movement-related sensor information to identify potential fall or abnormal activity events.

When a potential fall is detected, the system shall perform additional checks where possible, such as evaluating subsequent movement or user response, before escalating the event.

The system shall record detected events for further risk assessment and review.

FR14 – Medication Management

The system shall allow authorized users to configure medication information, including:

- Medicine name
- Dosage information where applicable
- Scheduled time
- Frequency
- Relevant instructions

The system shall maintain the medication schedule for each elderly user.

FR15 – Medication Reminder

The mobile application shall notify the elderly user when a scheduled medication is due.

The notification shall clearly communicate the relevant medication and scheduled time.

FR16 – Medication Confirmation

The elderly user shall be able to confirm whether a scheduled medication has been taken.

The system shall record the medication status, such as:

- Taken
- Missed
- Pending

The system shall store the corresponding time and date.

FR17 – Medication Adherence Analysis

The system shall calculate relevant medication-adherence information from recorded medication events.

The system shall make medication adherence information available as contextual information for health analysis and authorized review.

The system shall not automatically claim that a missed medication caused a subsequent health abnormality.

FR18 – Emergency and Risk Event Management

When the risk engine evaluates a Critical condition (such as a detected fall combined with abnormal vitals), the backend shall instantly dispatch push notifications to linked Caregiver smartphones.

FR19 – Elderly User Alert

The system shall notify the elderly user when an important warning or emergency condition is detected.

Where applicable, the system shall provide an opportunity for the elderly user to acknowledge or respond to the alert.

FR20 – Caregiver and Doctor Notification

Hospital clinic staff shall be able to scan the patient's mobile QR code without logging in to access a zero-login web dashboard displaying recent health vitals and trends, formatted with a CSS print stylesheet allowing 1-click Ctrl+P printing of a 1-page medical summary for the consulting doctor.

8.2 Non-Functional Requirements

NFR01 – Performance

The system shall provide timely responses for normal user interactions.

The mobile application should display locally available sensor information without unnecessary delay.

For backend operations, the system should provide an acceptable response time under the expected prototype workload.

For emergency notifications, the system shall minimize the time between confirmed emergency-event detection and notification generation.

Emergency-response latency will be measured during system evaluation.

NFR02 – Usability

The elderly mobile application shall use a simple and accessible interface appropriate for elderly users.

The interface should include:

- Clear labels
- Simple navigation
- Readable text
- Easily identifiable actions
- Minimal unnecessary interaction steps
- Clear notification messages
- Consistent interface elements

Usability will be evaluated using appropriate user-testing methods with representative or available test users.

NFR03 – Reliability

The system shall maintain reliable operation during normal monitoring activities.

The system shall:

- Validate incoming sensor data.
- Detect communication failures where possible.
- Prevent invalid measurements from being treated as valid health information.
- Maintain appropriate records of emergency events.
- Handle temporary connectivity failures gracefully.

The reliability of sensor communication and important workflows will be evaluated during testing.

NFR04 – Availability

The backend services should remain available during normal system operation.

The mobile application shall clearly indicate unavailable services or lost wearable connectivity rather than displaying stale data as if it were current.

The prototype's availability will be evaluated within the limitations of the selected hosting and development environment.

NFR05 – Security

The system shall protect user accounts and health-related information through appropriate security mechanisms.

The system shall include:

- Secure authentication.
- Role-based authorization.
- Secure API communication.
- Password protection using appropriate secure hashing mechanisms.
- Access restrictions for patient information.
- Protection against unauthorized modification of health and emergency records.
- Secure handling of authentication tokens or sessions.

Security testing will be performed against relevant application-level threats.

NFR06 – Privacy

The system shall collect and process only information required for the project's functions.

Health-related information shall only be accessible to authorized users according to their assigned roles and relationships.

The system shall avoid exposing sensitive health information in unnecessary notifications or interfaces.

Where location information is used for emergency assistance, it shall only be collected and shared for defined purposes and with appropriate authorization.

NFR07 – Maintainability

The system shall be designed using modular components with clear interfaces between the mobile application, backend, AI analysis, database, and IoT components.

The implementation shall follow consistent coding conventions and appropriate documentation practices.

The source code shall be maintained using GitHub with meaningful commits, branches, pull requests, and code reviews where applicable.

NFR08 – Scalability

The backend architecture should allow additional elderly users, wearable devices, health records, and authorized caregivers to be added without requiring fundamental changes to the system architecture.

The prototype will not target large-scale commercial deployment, but the design should avoid unnecessary architectural limitations.

NFR09 – Data Integrity

The system shall maintain the integrity and consistency of health measurements, medication records, risk assessments, and emergency events.

Important records shall contain appropriate timestamps and relationships to the relevant user or device.

The system shall prevent unauthorized modification or deletion of important health and emergency records.

NFR10 – Interoperability

The system shall use clearly defined interfaces between the mobile application, backend services, database, AI component, and wearable device.

The backend shall expose documented APIs for required mobile-system communication.

The wearable communication interface shall be designed so that supported sensors or compatible hardware can be replaced or extended without redesigning the entire system.

NFR11 – AI Model Performance

The AI-assisted analysis component shall be evaluated using appropriate measures based on the selected algorithm and available dataset.

Depending on the implemented approach, evaluation may include:

- Accuracy
- Precision
- Recall
- F1-score
- False-positive rate
- False-negative rate

The selected evaluation metrics shall correspond to the actual AI task implemented by the group.

NFR12 – Auditability and Traceability

Important system events shall be recorded with sufficient information to support analysis and debugging.

The system should maintain records for:

- Authentication events where appropriate.
- Health-data collection.
- Risk assessments.
- Medication events.
- Abnormal events.
- Emergency alerts.
- Notification status.

This information shall support system testing, troubleshooting, and evaluation.

NFR13 – Compatibility

The initial mobile application shall target supported Android devices and versions selected during implementation.

The system shall be tested on representative Android hardware and the selected wearable prototype.

NFR14 – Resource Efficiency

The mobile application and wearable prototype should use reasonable processing, memory, storage, battery, and network resources.

The wearable component should minimize unnecessary data transmission and processing where possible to support practical operation within the selected hardware limitations.

NFR15 – Safety and Responsible Operation

The system shall clearly communicate that its risk assessment and alerts are intended to provide assistance and decision support rather than medical diagnosis.

The system shall not autonomously prescribe treatment, modify medication, or make clinical decisions on behalf of healthcare professionals.

Limitations, false positives, false negatives, sensor errors, and connectivity failures shall be documented and considered during evaluation.

9. Proposed Solution

9.1 Overall Solution Concept

ElderGuard AI is an AI-powered IoT elderly health monitoring and assistance platform structured across a 5-tier architecture. It connects an ESP32 wearable sensor device, a dual-mode Flutter mobile app, a FastAPI backend with MySQL storage, an emergency alert dispatch service, and a zero-login hospital web dashboard for doctor printouts.

The wearable collects selected health and activity data and sends it to the mobile application through **BLE**. The mobile application sends the data to the backend, where it is stored and analysed. The AI component analyses current data together with historical data, medication adherence, and doctor-defined patient parameters. The risk-assessment module then classifies the user's condition as **Normal, Warning, High Risk, or Critical**.

9.2 Major Features

The proposed system will provide:

- IoT-based health and activity monitoring
- Elderly-friendly mobile health dashboard
- Medication reminders and confirmation
- Historical health-data management
- AI-assisted anomaly and pattern detection
- Personalized risk assessment

- Fall/emergency detection
- Caregiver and doctor notifications
- Health and medication trend analysis
- Role-based access control and security

9.3 System Workflow

IoT Wearable → Mobile App → Backend → Database → AI Analysis → Risk Assessment → Emergency Response → Notification

For example, if the system detects a fall together with abnormal health measurements and no user response, it can classify the event as **Critical** and notify the configured caregiver/doctor.

9.4 User Interaction

The system supports two main users:

User	Main Functions
Elderly Person	View health data, receive reminders/alerts, confirm medication
Caregiver/Family	Monitor elderly person and receive alerts

9.5 Major Inputs and Outputs

Inputs:

- Wearable sensor data
- Patient-specific health parameters
- Medication information
- Medication confirmation
- Historical health data
- Activity/fall events

Outputs:

- Health dashboard
- AI analysis results
- Risk level
- Medication reminders
- Warning/emergency notifications
- Health and medication reports

9.6 Component Integration & 5-Tier Architecture

The system is structured across 5 integrated tiers:

Layer 1 (ESP32 Wearable): Collects vital signs (heart rate, SpO2, body temperature) and motion/fall data, transmitting raw packets to the phone via Bluetooth Low Energy (BLE).

Layer 2 (Flutter Mobile App): Runs in Elderly Mode (displays vitals, handles local medication/food reminders, SOS button, displays unique QR code) or Caregiver Mode (remote vital views, alert reception).

Layer 3 (FastAPI Backend & MySQL Database): Receives telemetry via REST/HTTPS, stores historical measurements, and evaluates real-time risk status (Normal, Warning, Critical).

Layer 4 (Emergency Alert Service): Immediately dispatches push notifications to the Caregiver's smartphone upon Critical risk or SOS activation.

Layer 5 (Hospital Emergency Web Dashboard): Enables hospital clinic staff to scan the patient's QR code, open a zero-login web view, and press Ctrl+P to print a 1-page health summary for the consulting doctor.

9.7 High-Level Workflow

Figure 2.1: High-Level Workflow of ElderGuard AI

Elderly Person



IoT Wearable



Mobile Application



Backend + Database



AI Analysis



Risk Assessment



Normal / Warning / High Risk / Critical



Notification / Emergency Response

↓

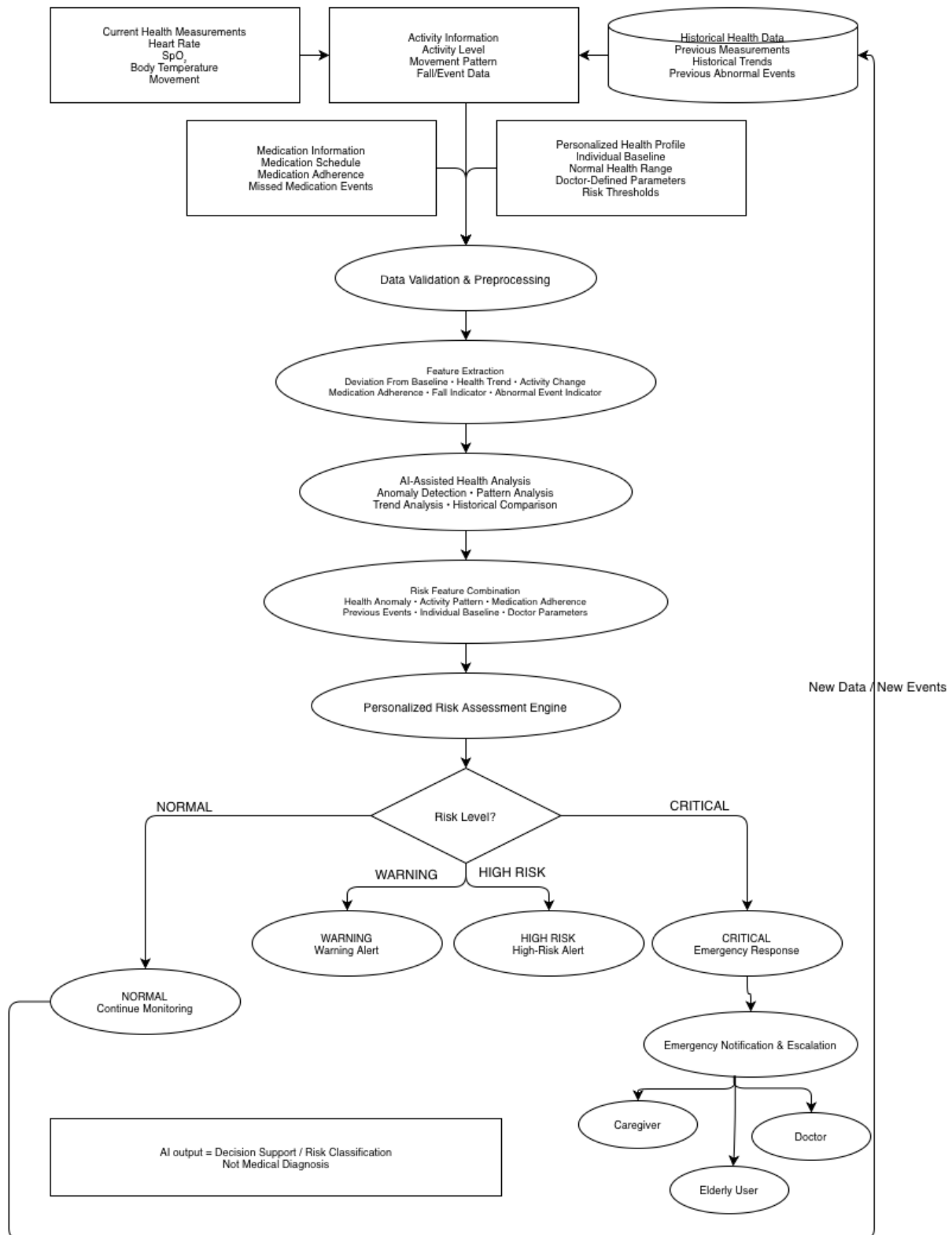
Caregiver / Elderly Person

The proposed solution therefore provides an integrated workflow:

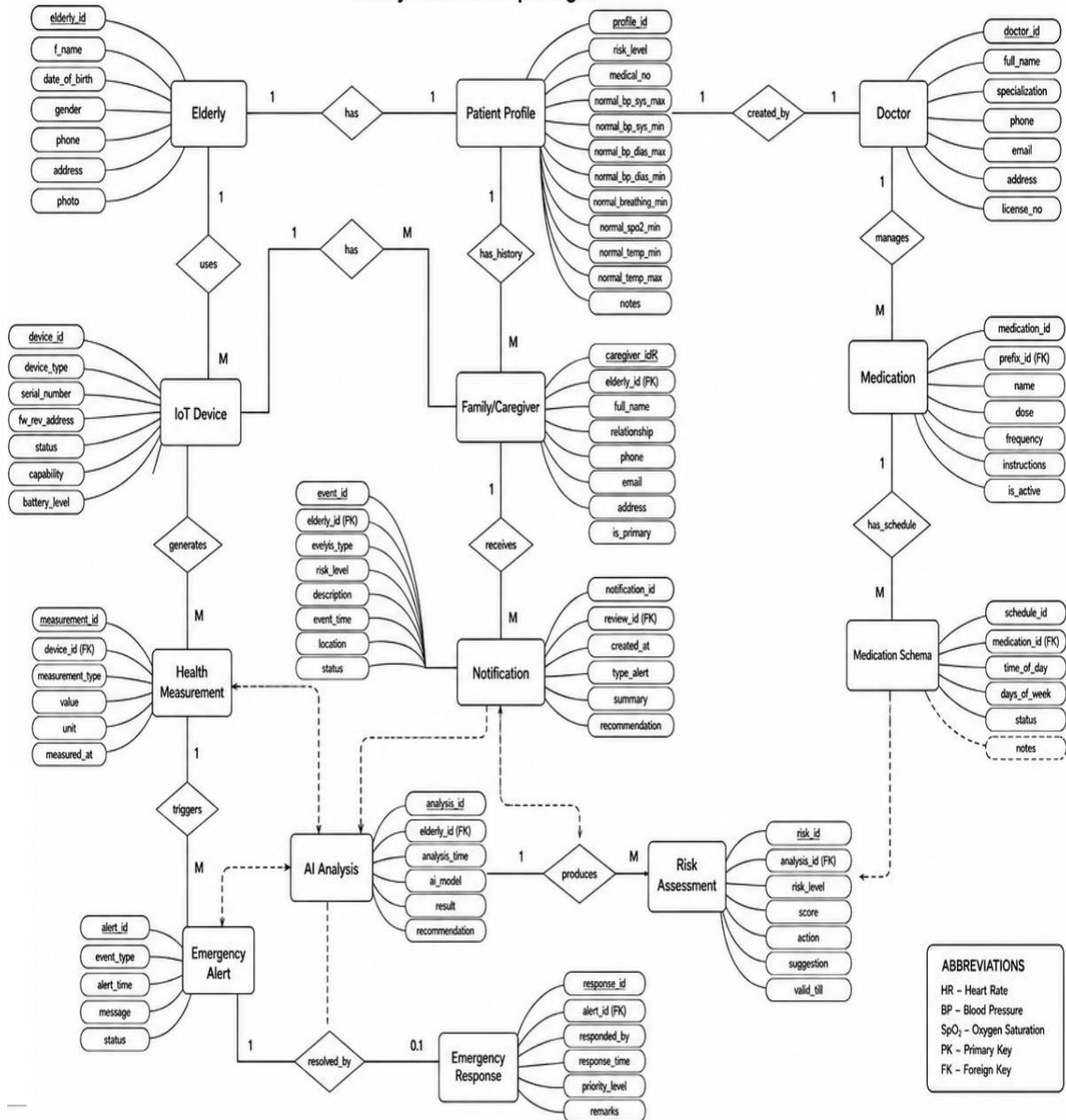
Monitor → Collect → Analyse → Assess → Respond → Record

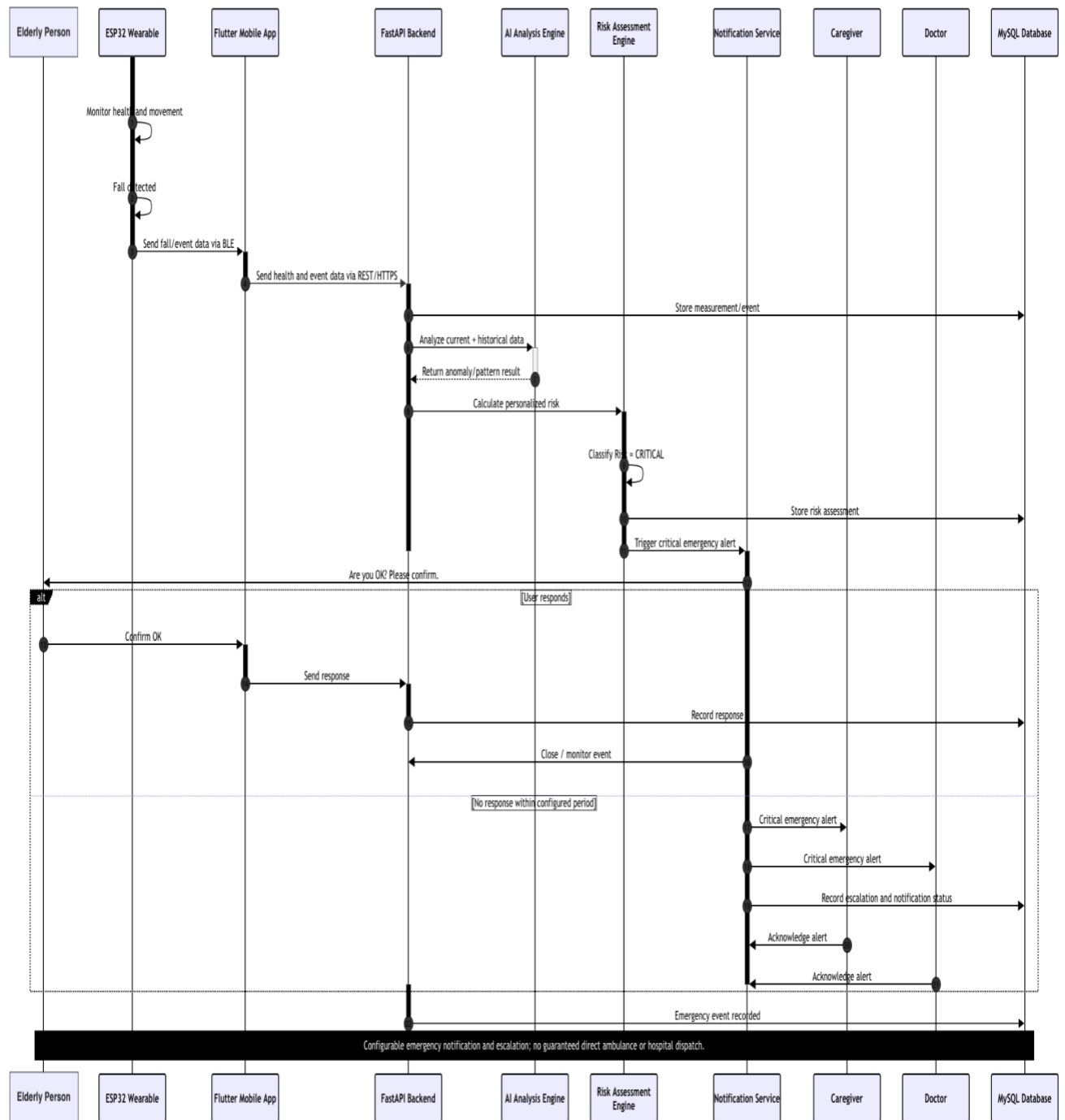
10. System Architecture and Technical Design

AI-Based Personalized Risk Assessment Architecture

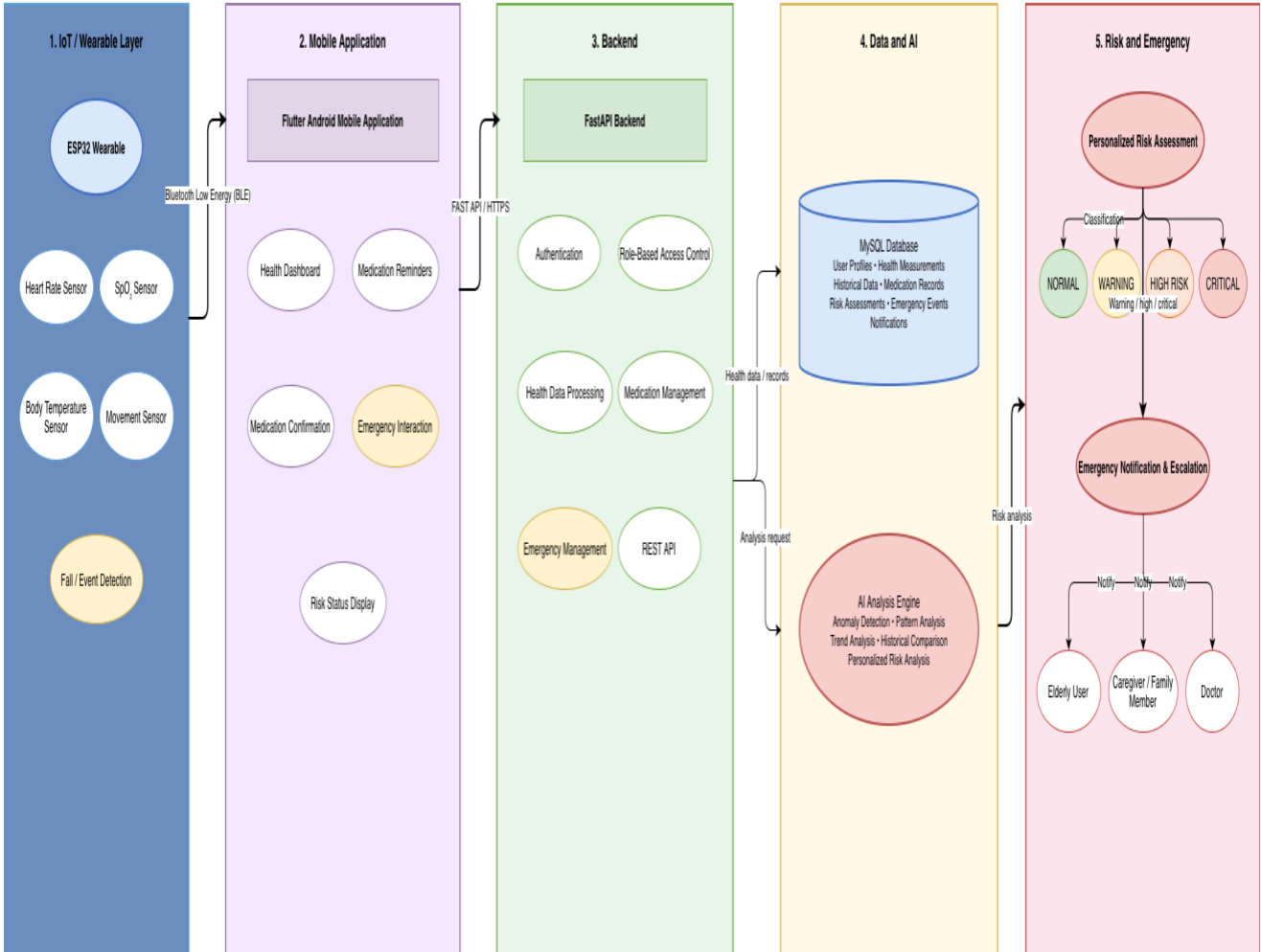


AI-Based Elderly Care & Emergency Assistance System Entity Relationship Diagram

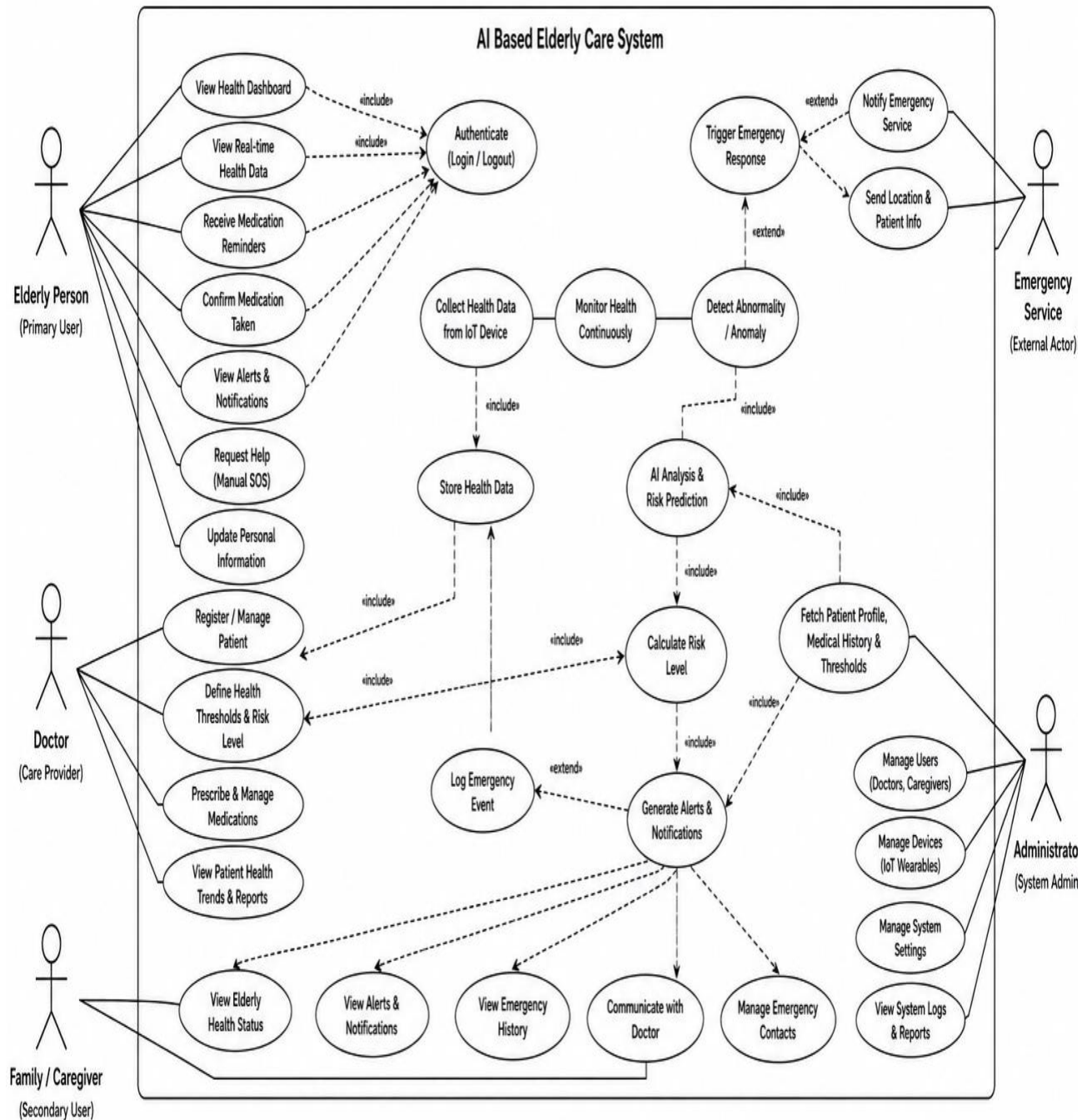




ElderGuard AI – Overall IoT / System Architecture



AI BASED ELDERLY CARE SYSTEM – USE CASE DIAGRAM



11. Technology Selection and Justification

The technology stack for **ElderGuard AI** is selected based on the project's requirements, mobile-app focus, IoT integration, AI processing, security, maintainability, team capability, and feasibility within the Mini Project period.

Category	Selected Technology	Purpose
Mobile Platform	Android & iOS	Cross-platform elderly mobile application
Mobile Framework	Flutter	Cross-platform mobile application development
Mobile Language	Dart	Mobile application development
Backend Language	Python	Backend and AI/ML development
Backend Framework	FastAPI	REST API and backend service development
Database	MySQL	Storage of user, health, medication and emergency data
IoT Hardware	ESP32	Wearable/IoT prototype controller
Sensors	Selected health and motion sensors	Health, activity and fall-related data collection
Communication	Bluetooth Low Energy (BLE)	Wearable-to-mobile communication
API Communication	REST + HTTPS	Secure mobile-to-backend communication
Authentication	JWT	User authentication and role-based access
Security	Password Hashing + RBAC	Protection of user and health-related data
Development Tools	VS Code	Software and mobile application development
API Testing	Postman	REST API development and testing
Version Control	Git + GitHub	Source-code management and collaboration
UI/UX Design	Figma	User-interface design and prototyping
Project Management	GitHub Projects	Task, milestone and progress management
Deployment	Cloud Platform	Backend, database and AI service deployment

12. Technical Depth and Core Engineering Work

ElderGuard AI will involve significant technical development beyond basic CRUD operations. The core engineering work will focus on integrating **mobile application development, IoT communication, AI-assisted analysis, personalized risk assessment, security, and real-time emergency response.**

12.1 Major Technical Work

- Develop a **cross-platform Flutter mobile application** for elderly users.
- Establish **BLE communication** between the ESP32-based wearable and mobile application.
- Implement reliable collection, validation, and transmission of health and activity data.
- Develop **FastAPI REST services** for secure communication between the mobile application and backend.
- Design and implement the relational **MySQL database** for health, medication, user, and emergency data.
- Develop AI-assisted **anomaly detection and pattern analysis** using historical and current health data.
- Implement **personalized risk assessment** using patient-specific and doctor-defined parameters.
- Develop medication-adherence tracking and integrate it with health analysis.
- Implement fall/emergency event detection and risk-level classification.
- Develop an **automated emergency notification and escalation workflow.**
- Implement JWT authentication and role-based access control.
- Develop health trends and historical analysis for authorized doctors and caregivers.

12.2 Major Technical Challenges and Proposed Approaches

Technical Challenge	Proposed Approach
Reliable wearable-to-mobile communication	Use ESP32 with BLE and implement connection/data validation mechanisms
Noisy or abnormal sensor data	Apply data validation and preprocessing before analysis
Personalized health assessment	Use doctor-defined parameters and individual historical baselines
Detecting unusual health patterns	Apply appropriate anomaly-detection/ML techniques
Combining multiple health indicators	Develop a risk-assessment mechanism using multiple input features
Real-time emergency detection	Process incoming measurements and event information with predefined response rules
Medication and health relationship analysis	Combine medication adherence history with health measurements as contextual features
Protecting health-related data	Implement JWT authentication, authorization,

	password hashing and HTTPS
Large amounts of historical data	Design appropriate database structures and data-processing methods
Cross-platform mobile and IoT integration	Implement BLE communication and platform-compatible Flutter components

12.3 Core Engineering Contribution

The main technical contribution of the project is the integration of:

IoT Data Collection → Mobile BLE Communication → REST Backend → Historical Data Processing → AI Analysis → Personalized Risk Assessment → Emergency Response

Therefore, the project will not be limited to database management or standard CRUD functionality. The **mobile application, IoT communication, AI-assisted analysis, and risk-assessment workflow** form the core engineering work of ElderGuard AI.

The project will be evaluated based on the correctness, reliability, performance, security, and effectiveness of these technical components.

13. Data, Security, Privacy and Ethical Considerations

ElderGuard AI processes health-related and personal information; therefore, **data security, privacy, user consent, and ethical handling of information** will be considered throughout the system design and implementation.

13.1 Data Types and Sources

The system may collect:

- User profile information
- Health measurements such as heart rate, SpO₂ and temperature
- Activity and movement data
- Fall-related events
- Medication schedules and adherence records
- Historical health records
- Risk-assessment results
- Emergency-event information
- Device information
- Location information during authorized emergency situations

Data will mainly originate from the **IoT wearable, mobile application, and authorized doctor/caregiver inputs**.

13.2 Authentication and Authorization

The system will implement:

- JWT-based authentication
- Secure password hashing
- Role-based access control
- Authenticated API access
- Permission-based access to patient information

Different access permissions will be provided for **elderly users, doctors, caregivers, and administrators**.

13.3 Data Protection

The following measures will be considered:

- HTTPS for communication between mobile application and backend
- Password hashing rather than storing plain-text passwords
- Input validation
- Secure API authentication
- Role-based authorization
- Restricted access to health information
- Secure database configuration
- Protection of sensitive configuration values and API credentials
- Appropriate logging without unnecessarily exposing sensitive information

13.4 Privacy and Consent

Users will be informed about the types of information collected and the purpose of data processing.

Consent will be considered for:

- Health-data collection
- Wearable monitoring
- Location access during emergency situations
- Sharing relevant information with authorized caregivers/doctors

Only information required for the intended functionality will be collected where practical.

13.5 Ethical Considerations

The system is intended to provide **decision support and emergency assistance**, not to replace professional medical diagnosis.

Therefore:

- AI predictions will not be presented as medical diagnoses.
- AI-generated risk results will be treated as supportive information.

- The system will avoid claiming that a medication caused a particular health event without medical evidence.
- Emergency decisions will be designed conservatively and transparently.
- Authorized users will only access information relevant to their responsibilities.
- Users should be informed about the limitations and possible errors of automated analysis.

13.6 Data Reliability and Safety

Sensor readings may contain errors or temporary communication failures. Therefore, the system will consider:

- Sensor-data validation
- Missing-data handling
- Communication failures
- Abnormal/out-of-range readings
- Timestamp validation
- Device connection status

The system should avoid treating a single potentially incorrect sensor reading as a confirmed medical emergency without considering other available information.

13.7 Ethical and Legal Responsibility

Since the project handles health-related information, the implementation will follow applicable **data-protection, privacy, and ethical requirements** relevant to the project's deployment context.

The prototype will clearly communicate that ElderGuard AI is an **academic decision-support and assistance system**, and that users should rely on qualified healthcare professionals and official emergency services for actual medical emergencies.

14. Testing and Evaluation Plan

ElderGuard AI will be evaluated through multiple levels of testing to verify the correctness, reliability, security, performance, and usability of the system.

14.1 Testing Approach

Testing Type	Purpose
Unit Testing	Test individual functions, services, AI components and mobile components
Integration Testing	Verify communication between wearable, mobile app, backend, database and AI components
System Testing	Verify the complete ElderGuard AI workflow

Hardware Testing	Verify ESP32 sensors, BLE communication and fall/event detection
API Testing	Verify REST API functionality, validation, authentication and authorization
AI Evaluation	Evaluate anomaly detection and risk-classification performance
Security Testing	Test authentication, authorization and protection of sensitive data
Performance Testing	Evaluate API response time, data processing and notification response
Usability Testing	Evaluate whether the mobile application is understandable and easy to use for elderly users

14.2 Functional Evaluation

The main system functions will be evaluated using test scenarios such as:

- Successful wearable-to-mobile data transmission
- Health-data storage and retrieval
- Medication reminder and confirmation
- Medication-missed detection
- AI anomaly detection
- Risk-level classification
- Fall/emergency event detection
- Emergency notification
- Doctor and caregiver access
- Role-based authorization

14.3 AI Evaluation

The AI component will be evaluated using appropriate metrics depending on the selected model and available dataset, such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Special attention will be given to **false negatives in critical-risk detection**, because failing to identify a potential emergency can have greater consequences than generating an additional warning.

14.4 Hardware and Communication Evaluation

The IoT component will be tested under different conditions to evaluate:

- Sensor-data accuracy and consistency

- BLE connection reliability
- Data transmission success
- Temporary connection loss and recovery
- Fall/event detection behaviour
- Device response during continuous monitoring

14.5 Performance Evaluation

The system will be evaluated for:

- Mobile application responsiveness
- REST API response time
- Health-data processing time
- AI-analysis processing time
- Emergency notification delay
- System behaviour under multiple data records

14.6 Security Evaluation

Security testing will include:

- Login/authentication testing
- JWT validation
- Role-based access testing
- Unauthorized API access attempts
- Input validation
- Password protection
- Secure communication verification

14.7 Usability Evaluation

The elderly mobile application will be evaluated based on:

- Ease of navigation
- Readability
- Button and text visibility
- Clarity of health information
- Ease of medication confirmation
- Understanding of alerts
- Number of steps required for important actions

Feedback from selected test users or representative users will be used to identify usability improvements.

14.8 Overall Evaluation

The final evaluation will compare the testing results against the project's stated objectives and requirements. Test cases, results, screenshots, logs, performance measurements, and relevant AI evaluation results will be maintained as evidence.

The overall objective is to demonstrate that ElderGuard AI can reliably:

Monitor → Collect → Analyse → Assess Risk → Respond → Record

while maintaining appropriate performance, security, usability, and reliability.

15. Development Process and GitHub Plan

The **ElderGuard AI** project will follow an **Agile and iterative development approach**. The system will be developed incrementally through manageable tasks, with regular testing, integration, and progress reviews.

Development Workflow

The proposed development workflow is:

Requirements Analysis → System Design → Implementation → Testing → Code Review → Integration → Evaluation

The project will be divided into modules and features, with each group member responsible for significant technical components.

GitHub Repository

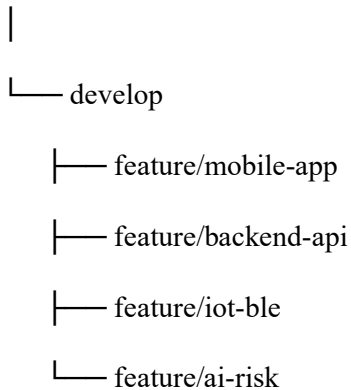
GitHub will be used as the official version-control platform throughout the project. The repository will contain source code, database scripts, API documentation, system diagrams, test-related artefacts, and other relevant project materials.

Each group member will use an **individually identifiable GitHub account** and contribute continuously throughout the project.

Branching Strategy

The team will use a feature-based branching strategy:

main



The **main** branch will contain stable versions, while **develop** will be used for integration. Feature branches will be created for individual features or modules.

Commit and Code Review Practices

Each member will make **regular and meaningful commits** related to their assigned work. Commit messages will clearly describe the implemented changes.

Completed features will be submitted through **Pull Requests** for review. Significant changes will be reviewed by another group member before being integrated into the shared development branch.

Integration Approach

The major components of ElderGuard AI, including the **Flutter mobile application, FastAPI backend, MySQL database, AI/ML component, and ESP32/BLE IoT component**, will be integrated incrementally.

API interfaces and data formats will be defined before integrating dependent components. Integration testing will be performed after major components are connected.

Project Management

GitHub Projects will be used to manage project tasks, milestones, issues, deadlines, and individual responsibilities.

Tasks will be tracked using stages such as:

To Do → In Progress → Review → Testing → Done

Individual Contribution

Each student will have primary ownership of at least one significant module and will participate in its **analysis, design, implementation, testing, integration, and technical documentation**.

Individual contributions will be reflected through GitHub commits, branches, Pull Requests, code reviews, and completed project tasks. Each member will be responsible for understanding and being able to explain the technical work attributed to them.

Repository Maintenance

The official GitHub repository will be maintained continuously throughout the implementation period. The repository link will be submitted to the Department/Mini Project Coordinator after proposal approval according to departmental requirements. Any required changes to the official repository after submission will be communicated to the Internal Supervisor/Mini Project Coordinator.

16. Project Management Plan

The **ElderGuard AI** project will follow a structured project management plan to ensure that development activities, responsibilities, milestones, integration, testing, and deployment are completed within the allocated project period.

GitHub Projects will be used as the primary project management and task-tracking tool. The project board will be continuously maintained throughout the implementation period.

16.1 Project Activities

The major project activities will include:

1. Requirements analysis and problem definition
2. System architecture and database design
3. UI/UX design and prototyping
4. Mobile application development
5. IoT and BLE communication development
6. Backend and REST API development
7. AI/ML and risk-assessment development
8. Module integration
9. Testing and debugging
10. Cloud deployment
11. Final evaluation and documentation

16.2 Project Schedule

The project will be developed through the following phases:

Phase	Major Activities	Responsible Members	Expected Output
Phase 1	Requirements analysis, literature/existing-solution study	All members	Requirements and gap analysis
Phase 2	Architecture, database and UI/UX design	All members	System designs and Figma prototype
Phase 3	Flutter mobile application development	All members	Mobile application modules
Phase 4	ESP32, sensors and BLE communication	All members	Working IoT prototype
Phase 5	FastAPI backend and REST API development	All members	Backend services and APIs
Phase 6	A and personalized risk assessment	All members	AI analysis and risk-assessment component
Phase 7	Module integration	All members	Integrated ElderGuard AI system
Phase 8	Functional, integration, security, performance and usability testing	All members	Test results and defect reports
Phase 9	Cloud deployment and final improvements	All members	Deployed system
Phase 10	Final documentation, demonstration and evaluation preparation	All members	Final project and presentation materials

16.3 Milestones

The major milestones will include:

- **M1:** Requirements and existing-solution analysis completed
- **M2:** System architecture and UI/UX design completed
- **M3:** Initial mobile and IoT components completed
- **M4:** Backend and database implementation completed
- **M5:** AI/risk-assessment component completed
- **M6:** Major system integration completed
- **M7:** Testing and debugging completed
- **M8:** Cloud deployment completed
- **M9:** Final evaluation and documentation completed

16.4 Task Management

GitHub Projects will be used to maintain:

- Project tasks and activities
- Assigned group members
- Task priorities
- Deadlines
- Task status
- Milestones
- Issues and bugs
- Module progress
- Completed work

Tasks will generally be tracked using:

To Do → In Progress → Review → Testing → Done

The project management board will be updated continuously to reflect the actual progress of the project.

16.5 Gantt Chart

A **Gantt chart** will be included in the proposal to present the planned project timeline, activities, responsible members, milestones, integration, testing, deployment, and final evaluation preparation.

Task / Activity	Weeks														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Project Planning & Requirements															
Problem analysis & objectives															
Requirements analysis															
Technology & feasibility study															
2. System Design															
System architecture															
Database design															
UI/UX design & prototyping															
3. IoT & Wearable Module															
ESP32 & sensor setup															
Sensor data acquisition															
BLE communication															
Fall/activity detection															
4. Backend & Database															
Backend/API development															
Database implementation															
Authentication & RBAC															
5. Mobile Application															
Flutter application development															
Health dashboard															
Medication management															
Notifications & emergency UI															
6. AI & Risk Assessment															
Data preprocessing															
Anomaly detection															
Personalized baseline															
Risk assessment engine															
7. Doctor/Caregiver Module															
Doctor functionality															
Caregiver monitoring															
Health trends & reports															
8. System Integration															
IoT ↔ Mobile integration															
Mobile ↔ Backend integration															
AI ↔ Risk assessment integration															
Emergency notification workflow															
9. Testing & Evaluation															
Unit & API testing															
Integration testing															
Hardware/BLE testing															
AI/ML evaluation															
Security & performance testing															
Usability testing															
10. Finalization & Presentation															
Bug fixing & optimization															
Documentation															
Final demo & presentation preparation															

17. Feasibility and Risk Analysis

The feasibility of **ElderGuard AI** has been considered in terms of technical requirements, available resources, project duration, cost, data availability, and external dependencies. The project is considered feasible as a Mini Project because the proposed system will be implemented as a **prototype focused on selected health-monitoring and emergency-assistance functions** rather than a complete clinical system.

17.1 Feasibility Analysis

Feasibility Area	Assessment
Technical Feasibility	The proposed technologies, including Flutter, FastAPI, MySQL, Python AI/ML libraries, ESP32, and BLE, are suitable for implementing the required system components.
Time Feasibility	The project will be divided into manageable modules and developed incrementally according to the project schedule. Core features will be prioritized before optional features
Resource Feasibility	The required development tools, software frameworks, GitHub, Figma, and development environments are accessible to the team.
Hardware Feasibility	ESP32 and selected sensors can be used to develop an affordable IoT prototype for health and activity monitoring.
Software Feasibility	The required development frameworks, libraries, testing tools, and database technologies are available for development.
Cost Feasibility	Open-source technologies will be prioritized. Hardware and cloud costs will be controlled by using affordable prototype components and suitable free or low-cost services where available.
Data Feasibility	Publicly available datasets and controlled/simulated sensor data may be used for AI development and evaluation where real patient data is unavailable.
Deployment Feasibility	The backend and related services can be deployed using suitable cloud infrastructure.
External Dependencies	The project may depend on sensor availability, cloud services, mobile permissions, internet connectivity, and third-party notification services. These dependencies will be considered during development.

17.2 Risk Analysis

Risk	Impact	Mitigation Strategy
IoT sensor availability or hardware failure	High	Use affordable and replaceable sensors and maintain simulated sensor data for software testing.
BLE connection instability	High	Implement connection monitoring, reconnection mechanisms, and data validation.
Inaccurate or noisy sensor data	High	Apply data validation, preprocessing, and filtering before analysis.
Insufficient AI training data	High	Use suitable public datasets and controlled/simulated data while clearly documenting dataset limitations.
AI produces incorrect risk classifications	High	Evaluate models using appropriate metrics and combine AI analysis with validated rule-based risk conditions where necessary.
Project scope becomes too large	High	Prioritize core features and treat advanced features as optional extensions.
Integration problems between modules	Medium	Define API contracts and data formats early and perform incremental integration testing.
Cloud service limitations or cost	Medium	Select suitable free/low-cost services and maintain a local development environment as a backup.
Security or unauthorized access	High	Implement JWT authentication, role-based authorization, password hashing, HTTPS, input validation, and secure configuration.
Mobile platform/device compatibility issues	Medium	Test the Flutter application on multiple supported devices and prioritize Android/iOS compatibility requirements.
Loss of development work	Medium	Use Git and GitHub continuously with regular commits and remote repository backups
Team member availability or workload issues	Medium	Use GitHub Projects to track responsibilities, deadlines, and progress, while maintaining knowledge sharing among members.
Emergency notification	High	Provide multiple notification mechanisms

failure		where feasible and clearly define the limitations of the prototype
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17.3 Overall Feasibility

The project is considered **technically and practically feasible** within the Mini Project period provided that the team maintains a controlled scope and prioritizes the core functions. The implementation will focus on demonstrating the integration of **IoT monitoring, cross-platform mobile development, AI-assisted risk assessment, secure backend services, and emergency assistance**.

18. Expected Outcomes and Deliverables

At the end of the project, **ElderGuard AI** is expected to provide a functional prototype that demonstrates intelligent elderly monitoring, personalized risk assessment, medication assistance, and emergency response.

18.1 Expected Outcomes

The project is expected to deliver:

- A **cross-platform Flutter mobile application** for elderly users, doctors, and caregivers.
- An **ESP32-based IoT prototype** capable of collecting selected health and activity data.
- **BLE communication** between the IoT device and mobile application.
- A **FastAPI-based REST backend** for managing users, health data, medications, devices, and emergency events.
- A **MySQL database** for storing structured health and system information.
- An **AI-assisted analysis component** for anomaly detection, pattern analysis, and personalized risk assessment.
- A **risk-classification mechanism** that categorizes situations into appropriate risk levels.
- A **medication reminder and adherence tracking system**.
- An **emergency detection and notification mechanism** for configured caregivers/doctors.
- **Health trend and historical analysis** for authorized users.
- A secure authentication and authorization mechanism using **JWT and role-based access control**.
- A **cloud-deployed backend/system prototype** for demonstrating remote access and operation.

18.2 Project Deliverables

The following artefacts will be produced:

Deliverable	Description
ElderGuard AI Mobile Application	Functional Flutter application for supported mobile platforms
IoT Prototype	ESP32 with selected health/activity sensors
BLE Communication Component	Communication between wearable prototype and mobile application
Backend System	FastAPI REST APIs and backend services
AI Component	Anomaly detection, pattern analysis and risk-assessment functionality
MySQL Database	Database schema, relationships and required data
System Design Artefacts	Architecture, use-case, sequence, ER and other relevant diagrams
UI/UX Prototype	Figma designs and application prototypes
Testing Evidence	Test cases, results, defect records and evaluation results
Source Code	Complete project source code maintained in GitHub
Project Management Records	GitHub Projects tasks, milestones, issues and progress records
Technical Documentation	Project documentation and relevant implementation details
Cloud Deployment	Deployed backend/services where technically and financially feasible
Final Demonstration	Demonstration of the integrated IoT → Mobile → Backend → AI → Risk → Emergency workflow

19. Use of Generative AI and AI-Assisted Development Tools

Generative AI and AI-assisted development tools may be used by the project team as **supporting tools** during the development of ElderGuard AI. Such tools may assist with learning, understanding technical concepts, code suggestions, debugging, documentation improvement, brainstorming, and identifying possible implementation approaches.

The team will remain fully responsible for the **accuracy, correctness, security, testing, and quality** of all submitted work. Any AI-assisted code or content will be reviewed, modified where necessary, tested, and understood by the responsible team member before being incorporated into the project.

20. References

- [1] World Health Organization, “Ageing and health,” World Health Organization, Oct. 1, 2025.
- [2] World Health Organization, “Falls,” World Health Organization, Apr. 26, 2021.
- [3] P. Bet, P. C. Castro, and M. A. Ponti, “Fall detection and fall risk assessment in older person using wearable sensors: A systematic review,” *International Journal of Medical Informatics*, vol. 130, Art. no. 103946, 2019, doi: 10.1016/j.ijmedinf.2019.08.006.
- [4] M. Schwenk *et al.*, “Performance and characteristics of wearable sensor systems discriminating and classifying older adults according to fall risk: A systematic review,” *Sensors*, 2021.
- [5] “Fall Detection in Elderly People: A Systematic Review of Ambient Assisted Living and Smart Home-Related Technology Performance,” *PubMed*, 2025.
- [6] S. Taborri *et al.*, “Wearable devices for improving medication adherence in older adults with chronic conditions: A systematic review,” *International Journal of Medical Informatics*, vol. 218, Art. no. 106515, 2026, doi: 10.1016/j.ijmedinf.2026.106515.
- [7] K. T. Haile *et al.*, “Digital health interventions to improve medication adherence among older adults: A systematic review,” *Australasian Journal on Ageing*, vol. 45, no. 3, Art. no. e70206, 2026, doi: 10.1111/ajag.70206.
- [8] Y.-H. Yap *et al.*, “Systematic review of the barriers affecting medication adherence in older adults,” *Geriatrics & Gerontology International*, 2015, doi: 10.1111/ggi.12616.
- [9] D. C. Das *et al.*, “The role of artificial intelligence in medication management for older adults: A systematic review,” *Aging Medicine*, vol. 9, no
- [10] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. 2008 Eighth IEEE International Conference on Data Mining*, 2008, pp. 413–422. . 2, pp. 196–213, 2026, doi: 10.1002/agm2.70074.
- [11] Espressif Systems, “Bluetooth Low Energy,” *ESP-IDF Programming Guide*, Espressif Systems.